一、

1. D. $AB=\phi$. $\therefore B\subset\bar{A}$. $P(\bar{A}\cup B)=P(\bar{A})=1-P$

2. D. 全概率公式. $P=\frac{3}{6}\times\frac{1}{10}+\frac{2}{6}\times\frac{1}{15}+\frac{1}{6}\times\frac{1}{20}=\frac{29}{360}$

3. B. $X_i\sim N(0,\sigma^2)$, 则 $\frac{X_i}{\sigma}\sim N(0,1)$. $V=\left(\frac{X_1}{\sigma}\right)^2+\left(\frac{X_2}{\sigma}\right)^2+\left(\frac{X_3}{\sigma}\right)^2\sim\chi^2(3)$

U与V独立. $\therefore\frac{U}{\sqrt{V/3}}\sim t(3)$ (t分布定义)

4. A. $D(Z)=D(X+2Y)=D(X)+4D(Y)+2\times1\times2\,Cov(X,Y)$

$=2^2+4\times3^2+2\times2\times\rho_{XY}\sqrt{D(X)}\sqrt{D(Y)}=52$

"0.5×2×3

5. C 双边检验，拒绝域 $|t|=\left|\frac{\bar{X}-\mu_0}{1.5/\sqrt{36}}\right|\geq t_{\frac{\alpha}{2}}(n-1)=2.030$

$\because|t|=0.6<t_{0.025}(35)$

$\therefore$ 接受 H_0.

6. D. $P\{0<X<12\}=P\{|X-6|<6\}\geq1-\frac{D(X)}{6^2}=\frac{5}{6}$

切比雪夫不等式.

二、1. 0 $f(x)=Ae^{-2x^2}=Ae^{-\frac{(x-0)^2}{2\times\frac{1}{4}}}$ 正态分布密度. $E(X)=0$.

2. $\Phi(2)$ $X_i\sim\pi(1)$. $E(X_i)=1$, $D(X_i)=1$.

$\sum_{i=1}^{100}X_i\xrightarrow{\text{近似}}N(100\times1,100\times1)$.

$=P\left\{\frac{\sum_{i=1}^{100}X_i-100}{10}<\frac{120-100}{10}\right\}\approx\Phi(2)$

3. $(2000\pm5t_{0.025}(8))$ $n=9$, μ, σ^2 都未知. $\left(\bar{X}\pm\frac{S}{\sqrt{n}}t_{\frac{\alpha}{2}}(n-1)\right)$

注意写清已知参数.

4. 10, 8.75. 设80人中患病人数为X, 则 $X\sim b(80,0.125)$

$E(X)=80\times0.125=10$, $D(X)=80\times0.125\times0.875=8.75$

5. 1, $\frac{1}{2}$. $\bar{x}=\frac{1}{5}(0+1+2+1+1)=1$. $S^2=\frac{1}{4}(1^2+0^2+1^2+0^2+0^2)=\frac{1}{2}$

注意 S^2 的系数.

6. 0. $\because f(x,y)=f_X(x)\cdot f_Y(y)$ X与Y独立.

$\therefore\rho_{XY}=0$.

三. 1. 设 B_i 表示取到第 i 个盒子. $i=1,2,3$.

A_i 表示第 i 次取到白球. $i=1,2$. 则

$P(B_i)=\frac{1}{3}$. $P(A_1|B_1)=\frac{3}{10}$. $P(A_1|B_2)=\frac{7}{15}$. $P(A_1|B_3)=\frac{5}{25}$.

(1) 由全概率公式.

$$P(A_1)=\sum_{i=1}^{3}P(B_i)P(A_1|B_i)=\frac{1}{3}\times\frac{3}{10}+\frac{1}{3}\times\frac{7}{15}+\frac{1}{3}\times\frac{5}{25}=\frac{29}{90}.$$

(2). $P(A_1|\bar{A}_2)=\dfrac{P(A_1\bar{A}_2)}{P(\bar{A}_2)}$

其中 $P(\bar{A}_2)=\sum_{i=1}^{3}P(B_i)\cdot P(\bar{A}_2|B_i)=\frac{1}{3}\times\frac{7}{10}+\frac{1}{3}\times\frac{8}{15}+\frac{1}{3}\times\frac{20}{25}=\frac{61}{90}$

$$P(A_1\bar{A}_2)=\sum_{i=1}^{3}P(B_i)\cdot P(A_1\bar{A}_2|B_i)=\frac{1}{3}\times\frac{3}{10}\times\frac{7}{9}+\frac{1}{3}\times\frac{7}{15}\times\frac{8}{14}+\frac{1}{3}\times\frac{5}{25}\times\frac{20}{24}=\frac{2}{9}$$

$$\therefore P(A_1|\bar{A}_2)=\frac{P(A_1\bar{A}_2)}{P(\bar{A}_2)}=\frac{20}{61}.$$

2. (1).

$$f_X(x)=\int_{-\infty}^{\infty}f(x,y)dy=\begin{cases}\int_0^1(x+y)dy=x+\frac{1}{2}, & 0<x<1\\ 0, & \text{其他}.\end{cases}$$

$$f_Y(y)=\int_{-\infty}^{\infty}f(x,y)dx=\begin{cases}\int_0^1(x+y)dx=\frac{1}{2}+y, & 0<y<1\\ 0, & \text{其他}.\end{cases}$$

(2). $\because f_X(x)\cdot f_Y(y)=\begin{cases}(x+\frac{1}{2})(\frac{1}{2}+y), & 0<x<1,0<y<1\\ 0, & \text{其他}\end{cases}$

$f(x,y)\neq f_X(x)\cdot f_Y(y)$

$\therefore$ X与Y不独立.

(3) $E(X)=\int_{-\infty}^{\infty}x\cdot f_X(x)dx=\int_0^1x(x+\frac{1}{2})dx=\frac{7}{12}$.

$E(X^2)=\int_0^1x^2\cdot(x+\frac{1}{2})dx=\frac{5}{12}$.

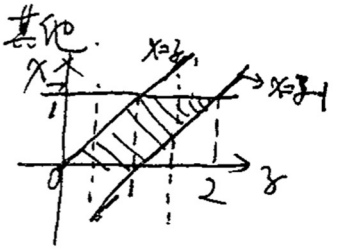
$D(X)=E(X^2)-[E(X)]^2=\frac{11}{144}$

(4). $f_{X|Y}(x|\frac{1}{4}) = \frac{f(x,\frac{1}{4})}{f_Y(\frac{1}{4})} = \begin{cases} \frac{x+\frac{1}{4}}{\frac{1}{2}+\frac{1}{4}} = \frac{4}{3}x+\frac{1}{3}, & 0<x<1 \\ 0, & \text{其他} \end{cases}$

$\therefore P\{X<\frac{1}{2} \mid Y=\frac{1}{4}\} = \int_{-\infty}^{\frac{1}{2}} f_{X|Y}(x|\frac{1}{4})dx = \int_0^{\frac{1}{2}} (\frac{4}{3}x+\frac{1}{3})dx = \frac{1}{3}$.

(5)

$$f(x,z-x) = \begin{cases} x+(z-x), & 0<x<1, 0<z-x<1 \\ 0, & \text{其他} \end{cases}$$



$Z = X+Y$

$f_Z(z) = \int_{-\infty}^{+\infty} f(x,z-x)dx$

$$= \begin{cases} \int_0^z z\,dx, & 0<z\le 1 \\ \int_{z-1}^1 z\,dx, & 1<z<2 \\ 0 & \text{其他} \end{cases} = \begin{cases} z^2, & 0<z\le 1 \\ z(2-z), & 1<z<2 \\ 0, & \text{其他} \end{cases}$$

3. 设 X_i 表示第 i 人身高. $X_i \sim N(178, 10^2)$, 且相互独立

$P\{X_1>180, X_2>180, \cdots, X_5>180\} = P\{X_1>180\}P\{X_2>180\}\cdots P\{X_5>180\}$

$= \prod_{i=1}^{5} P\{\frac{X_i-178}{10} > \frac{180-178}{10}\} = \prod_{i=1}^{5}[1-\Phi(\frac{1}{5})] = [1-\Phi(\frac{1}{5})]^5$.

4. (1) $f(x) = \begin{cases} \frac{1}{a}, & 0<x\le a \\ 0, & \text{其他} \end{cases}$

设样本值为 $x_1, \cdots, x_n$ 时似然函数为

$L(a) = \begin{cases} \frac{1}{a^n}, & \text{当 } x_1, \cdots, x_n \le a \text{ 即 } a \ge x_{(n)} = \max_{1\le i\le n}\{x_i\} \text{ 时.} \\ 0, & \text{否则} \end{cases}$

$\because$ 当 $a \ge \max_{1\le i\le n}\{x_i\}$ 时 $L(a)$ 单调递减.

$\therefore$ a 的最大似然估计是 $\hat{a} = \max\{X_1, \cdots, X_n\}$.

(2)

$$F(x)=\begin{cases}0, & x<0\\ \frac{x}{a}, & 0\le x<a\\ 1, & x\ge a\end{cases}$$

$$F_{\hat{a}}(x)=[F(x)]^n=\begin{cases}0, & x<0\\ \frac{x^n}{a^n}, & 0\le x<a\\ 1, & x\ge a\end{cases}$$

$$f_{\hat{a}}(x)=\begin{cases}\frac{n}{a^n}x^{n-1}, & 0<x<a\\ 0, & \text{其他}\end{cases}$$

$$E(\hat{a})=\int_{-\infty}^{+\infty}x f_{\hat{a}}(x)dx=\int_0^a x\cdot\frac{n}{a^n}x^{n-1}dx=\frac{n}{n+1}a\neq a$$

∴ $\hat{a}$ 不是 a 的无偏估计.

5. (1). $H_0:\sigma^2>0.21$. $H_1:\sigma^2<0.21$

检验统计量 $\chi^2=\frac{(n-1)S^2}{\sigma_0^2}=\frac{(n-1)S^2}{0.21}$.

拒绝域: $\chi^2\le\chi^2_{1-\alpha}(n-1)$.

$\alpha=0.05$. $n=46$. $\chi^2_{1-\alpha}(n-1)=\chi^2_{0.95}(45)=30.612$

∵ $\frac{(n-1)S^2}{\sigma_0^2}=\frac{45\times0.3}{0.21}=64.3>\chi^2_{0.95}(45)$

∴ 不拒绝 H_0. 即不接受这批零部.

(2). 枢轴量 $\chi^2=\frac{(n-1)S^2}{\sigma^2}\sim\chi^2(n-1)$ 即 $\chi^2(45)$

置信区间 $\left(\frac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2}}(n-1)},\frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2}}(n-1)}\right)$ 即 $(0.206,0.476)$

$1-\alpha=0.95$

$n=46$. $S^2=0.3$.

6. $X_1,\cdots,X_8$ 相互独立. 且 $X_i\sim N(0,1)$ $i=1,\cdots,8$

∴ $X_1+X_2+X_3+X_4\sim N(0,4)$. $\frac{X_1+X_2+X_3+X_4}{2}\sim N(0,1)$

同理 $\frac{X_5+X_6+X_7+X_8}{2}\sim N(0,1)$ 且相互独立.

∴ $\left(\frac{X_1+\cdots+X_4}{2}\right)^2+\left(\frac{X_5+\cdots+X_8}{2}\right)^2\sim\chi^2(2)$, $C=\frac{1}{4}$.